Title: FEA Project: Effects of Topology Optimization on Structural Strength
Name: Mihai Stanciu
Date: April 30, 2026

1 Introduction

Reducing structural weight while maintaining mechanical performance is a fundamental objective in engineering design. Topology optimization is a powerful tool for achieving this goal by systematically removing material while preserving load-carrying capacity. However, an important question arises: *How much mass can be removed through topology optimization before compressive structural performance begins to significantly degrade?*

This question is critical because many applications require lightweight components, but excessive mass reduction through topology optimization can degrade compressive strength and stability. Identifying this threshold is essential for ensuring reliable designs.

2 Methods

To answer the question, we employ the following steps:

1. Define the FEA workflow.
2. Create a simple geometry, such as a square pillar.
3. Create the compression model, perform mesh convergence, and select an appropriate mesh size.
4. Perform topology optimization with increasing mass reductions between 30% and 70% while applying a 3500 N compressive design load to the column.
5. Analyze the directional deformation and equivalent stress for the optimized geometries in Ansys for a range of loads varying from 0 to 10000 N in increments of 500 N using PLA as the material.
6. 3D print the geometries (using the same printer parameters such as infill, wall loops, and layer height) and test them under compression using a materials testing machine.
7. Visualize and compare the FEA and physical force displacement curves and analyze stress-strain curves from the FEA analysis.
8. Calculate the percent error between the predicted and experimental failure loads for each design.

2.1 FEA Workflow

Fig. 2 shows the Ansys workflow used to conduct the study. First, static structural is used for mesh convergence and a 3500 N compressive load is applied to the pillar as the basis for the topology optimization. Then, static structural is used for validating each of the optimized designs.

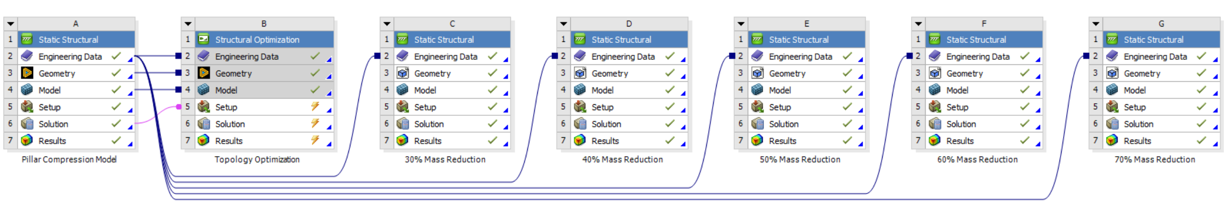

Figure 1: Ansys workflow showcasing the root static structural used for the structural optimization and the subsequent stress and deformation studies performed on each of the 5 optimized designs.

2.2 Geometry

For this study, we will use a simple rectangular pillar geometry that will facilitate mesh construction (no need for refinement or sizing), FEA computation time, and reduce 3D printing difficulty.

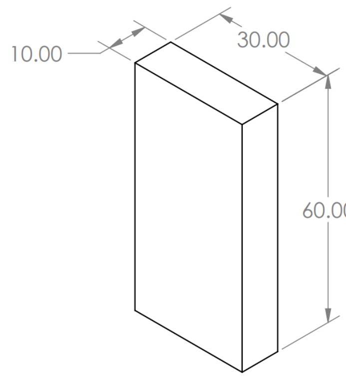

Figure 2: Geometry of the pillar used in the study. All dimensions are in millimeters.

2.3 Compression Model and Mesh Convergence

First, we create a simple static structural model for the pillar, shown in Fig. 3. The material used is PLA with the following properties:

$$\nu = 0.35, \quad E = 2.46\,\text{GPa}, \quad \rho = 1.32\,\text{g/cm}^3$$

For the boundary conditions, the pillar is fixed on the bottom face and a -3500 N compressive design load (Y-direction) is applied to the top face. This loading condition is used to perform the mesh convergence. A quick hand calculation yields an axial stress of 11.66 MPa caused by this compressive force.

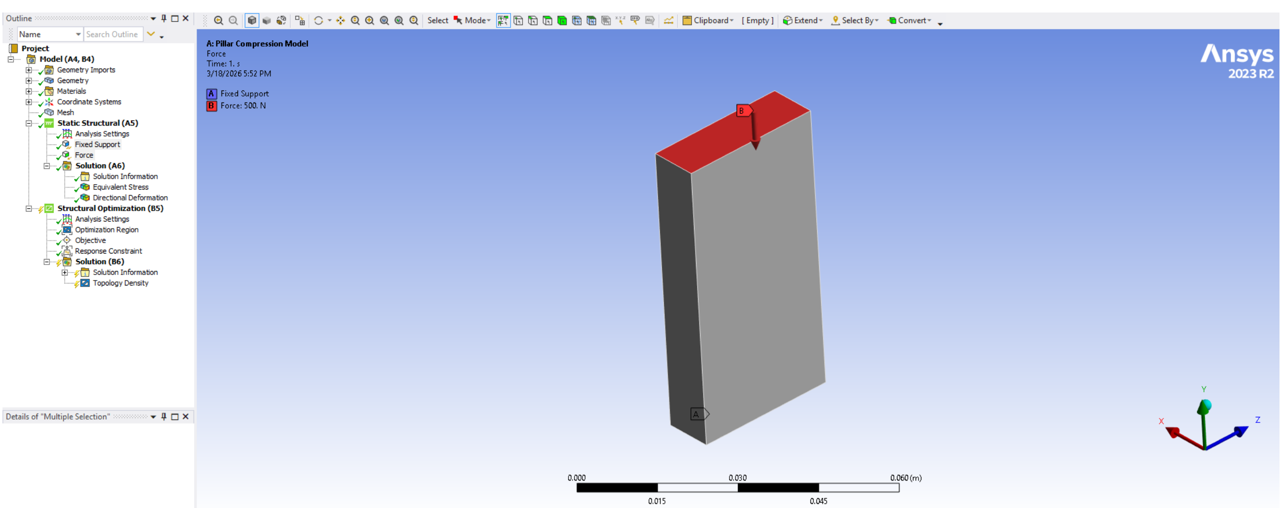

Figure 3: Static structural compression model for the pillar.

To optimize computational time and FEA results accuracy, we perform mesh convergence on the geometry presented in the previous section. From Fig. 4, we see that the mesh converges at around

11.485 MPa (average Von-Mises). Although we could select the mesh with 144 nodes (element size of 5.0 mm) since it is within 2% of the converged stress value, we select the model with 1152 nodes (element size of 2.5 mm) so that the topology optimized mesh is fine enough to produce smooth surfaces.

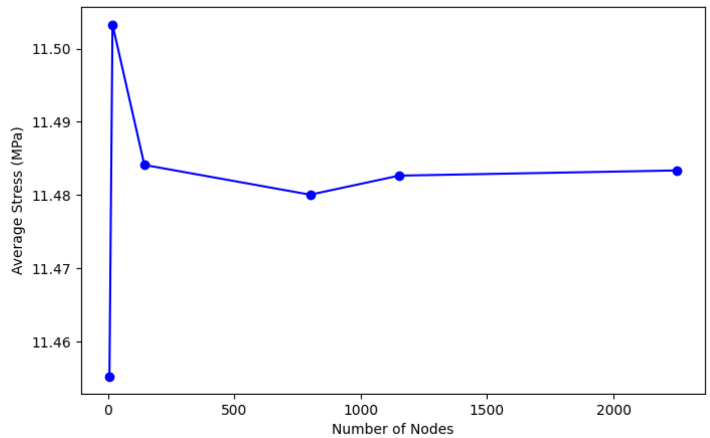

Figure 4: Mesh convergence graph for the rectangular pillar geometry.

We will use the same element size (2.5 mm) for creating the meshes for the topology optimized designs.

2.4 Topology Optimization

Next, we set up the structural optimization model to be able to use topology optimization. Fig. 5 shows the details of the model.

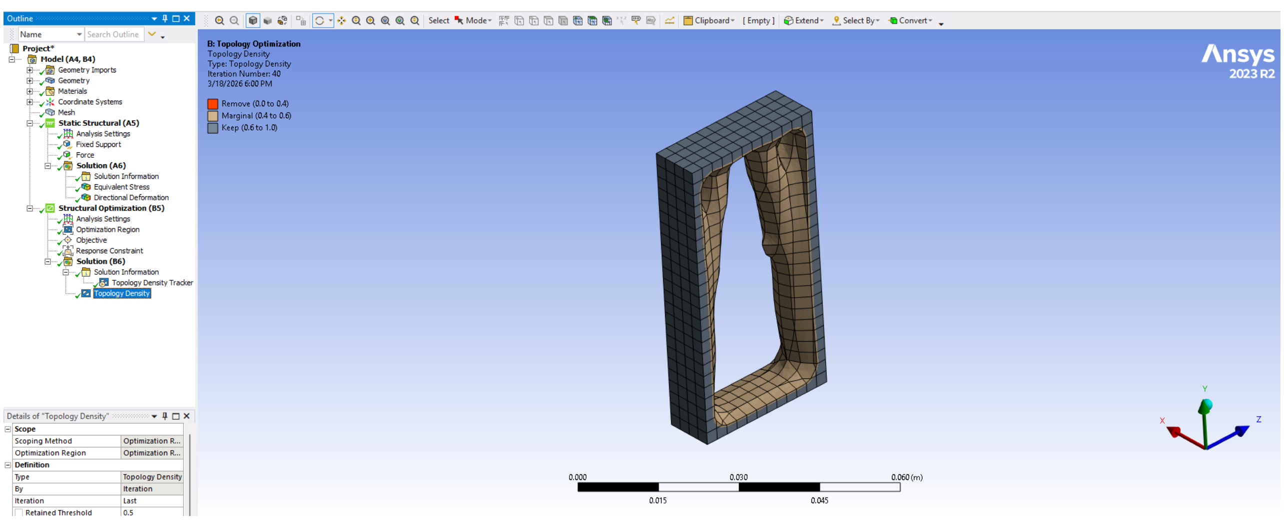

Figure 5: Structural optimization model for the pillar (70% mass reduction for this particular figure).

This process was repeated for mass reductions between 30 and 70% yielding the 5 topologies presented in Fig. 6.

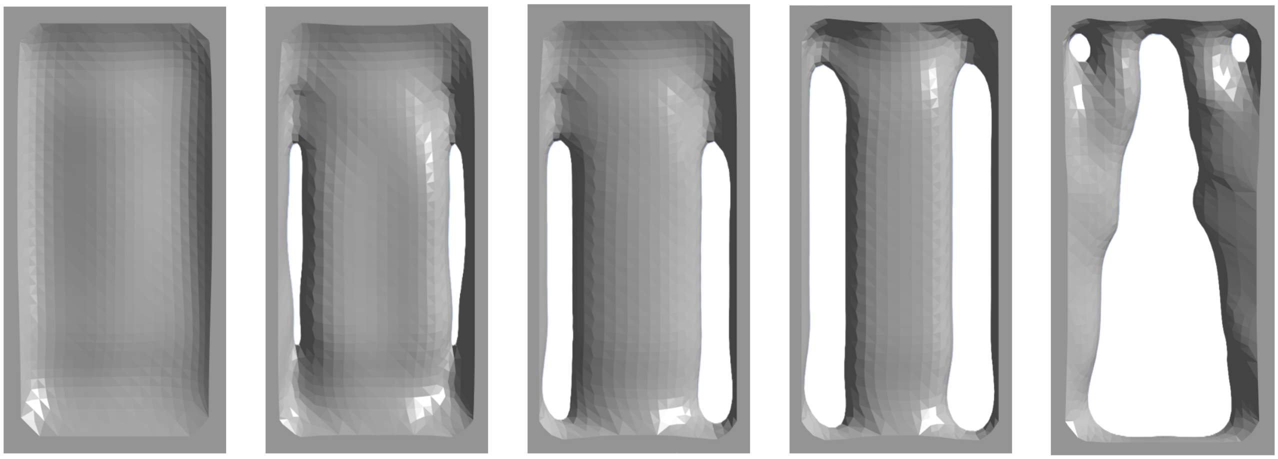

Figure 6: Five optimized topologies (lined up from left to right in increasing mass reduction order from 30% to 70% reduction).

2.5 Optimized Design Validation

We now validate the 5 topologies using the STL meshes from the topology optimization performed in the previous step. To do this, we must first create a patch-independent mesh using tetrahedra (this is native to STL meshes). Once the mesh is created, we must create named selections using the nodes on either face and the worksheet tool. The process is shown in Fig. 7 and is performed for the top and bottom faces to be able to set boundary conditions since the mesh does not retain those faces as selectable geometries. This study only uses small deflections since using large deflections would require a lot more time to set up.

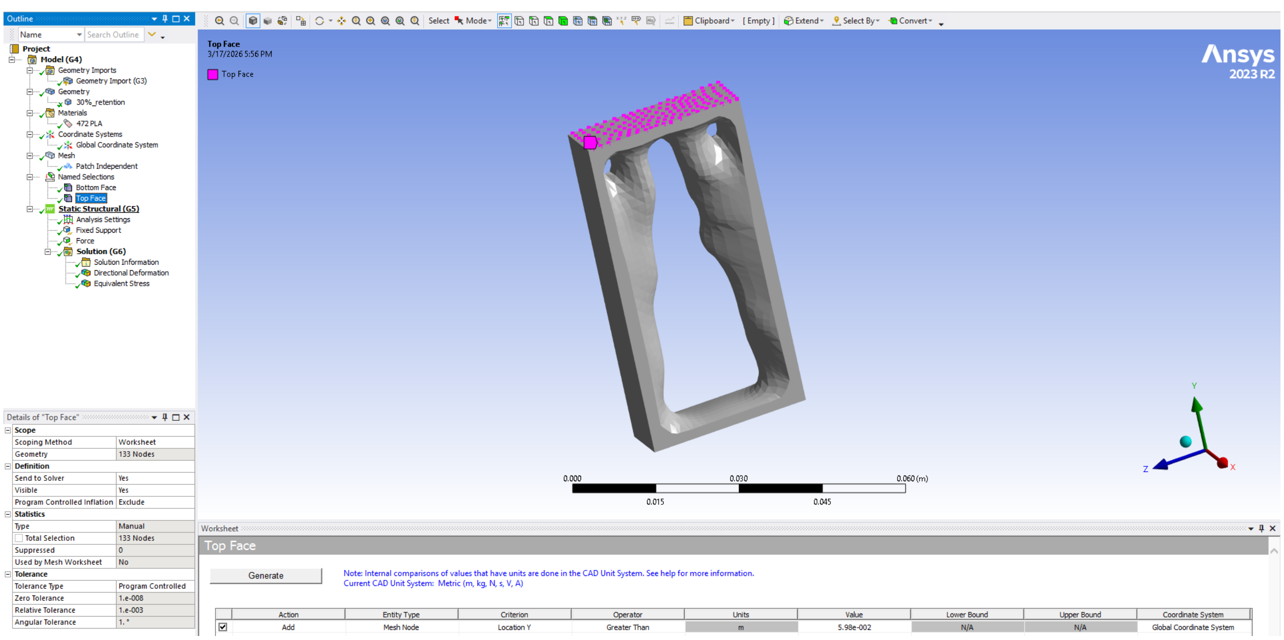

Figure 7: Named selection creation as a surface composed of nodes found on the top face. Same process is used for the bottom face.

Once both named selections are created and the boundary conditions are set, we solve for the directional deformation of the top face and the average equivalent Von-Mises stress due to loads ranging between 0 and 10000 N (the goal of such a large force is to capture yielding so that the validation may be compared to the physical tests). One of the results can be observed in Fig. 8.

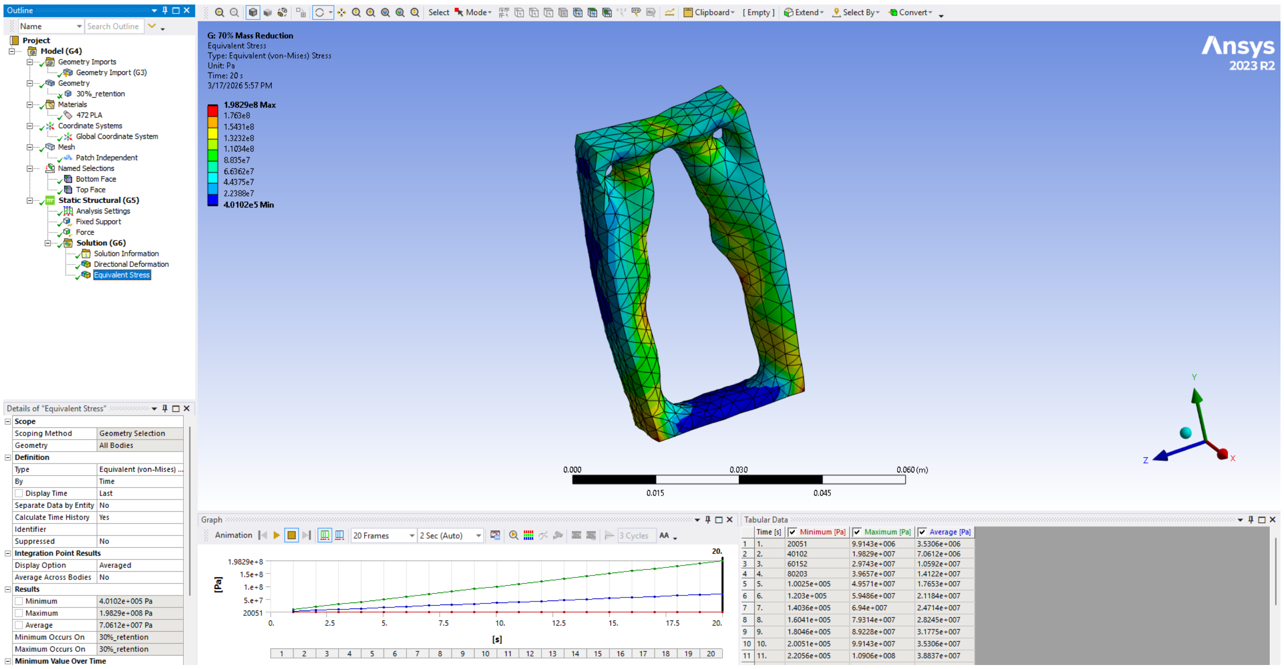

Figure 8: Average equivalent stress on the 30% optimized design.

2.6 Manufacturing the Specimens and Testing

Two sets of the 5 specimens were 3D printed using 100% infill, a layer height of 0.1 mm, out of PLA basic on a Bambu P1P machine. As shown in Fig. 9, the specimens were printed in two orientations, long and short grain, to observe the effects of FDM layers and to have multiple sets of data to compare the FEA model with.

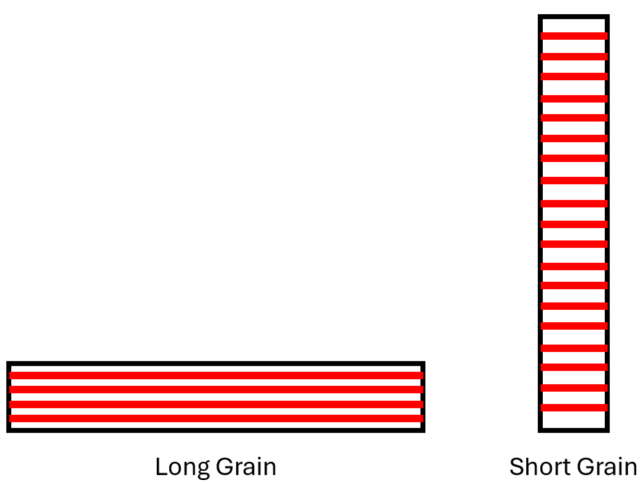

Figure 9: Two sets of topology specimens were 3D printed with different layer orientations.

Once manufactured, the specimens were tested in the 20kip 809 A/T Tensile Tester in compression.

3 Results

3.1 Force-Displacement Behavior

Fig. 10 shows the FEA model against the data collected during the experiments, with the maximum load of each specimen marked.

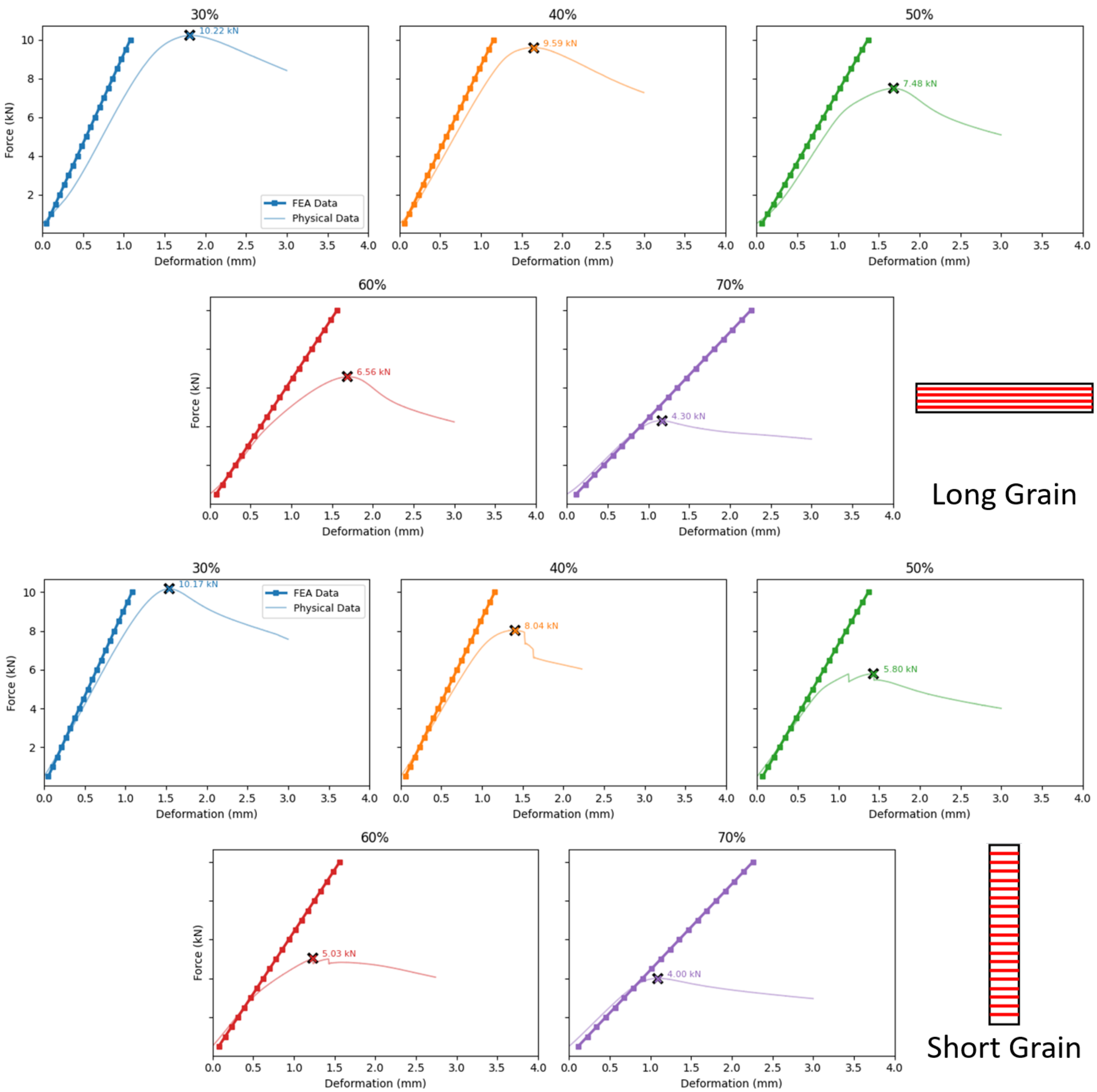

Figure 10: Force-displacement comparison between the FEA model and the physical data for the long and short grain specimens (all plots are graphed on the same scale)

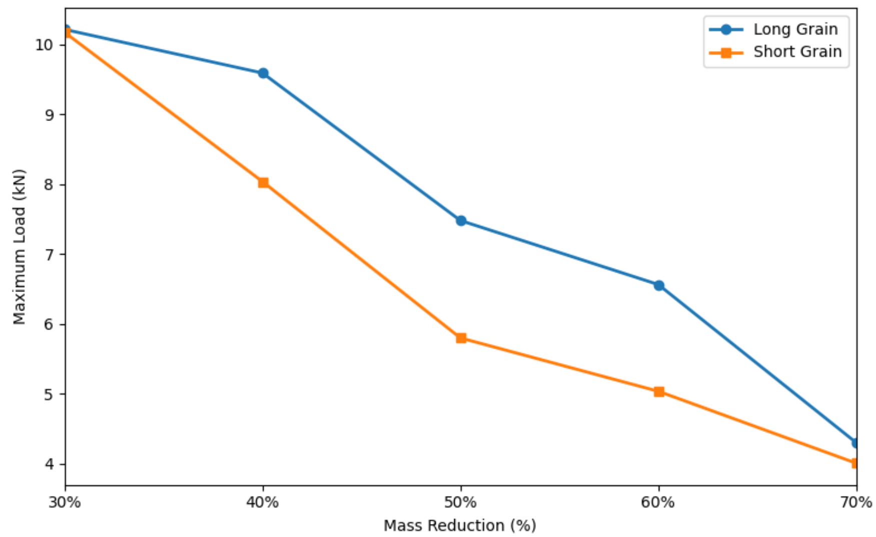

Figure 11: Comparison between the maximum strengths of the long and short grain specimens

3.2 FEA Stress Analysis

Since the FEA data agrees well with the physical data at our intended design load of 3500 N for each specimen, we plot the average stress of each sample at this load and mark the safety factor of each data point against a yield strength of 35 MPa. The trend is shown in Fig. 12.

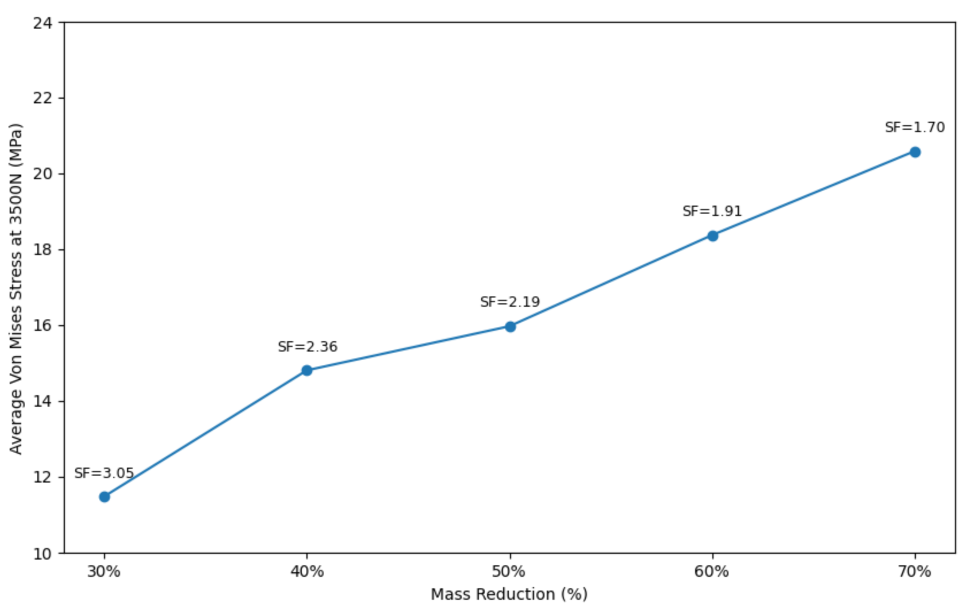

Figure 12: FEA-calculated average Von-Mises stress at the target design load of 3500N for each specimen with safety factors against yield strength annotated

3.3 Failure Mode

As expected, Fig. 12 shows buckling in each specimen; however, the short-grain specimens also experienced some cracking due to layer delamination. This delamination also caused all slender members to completely fail.

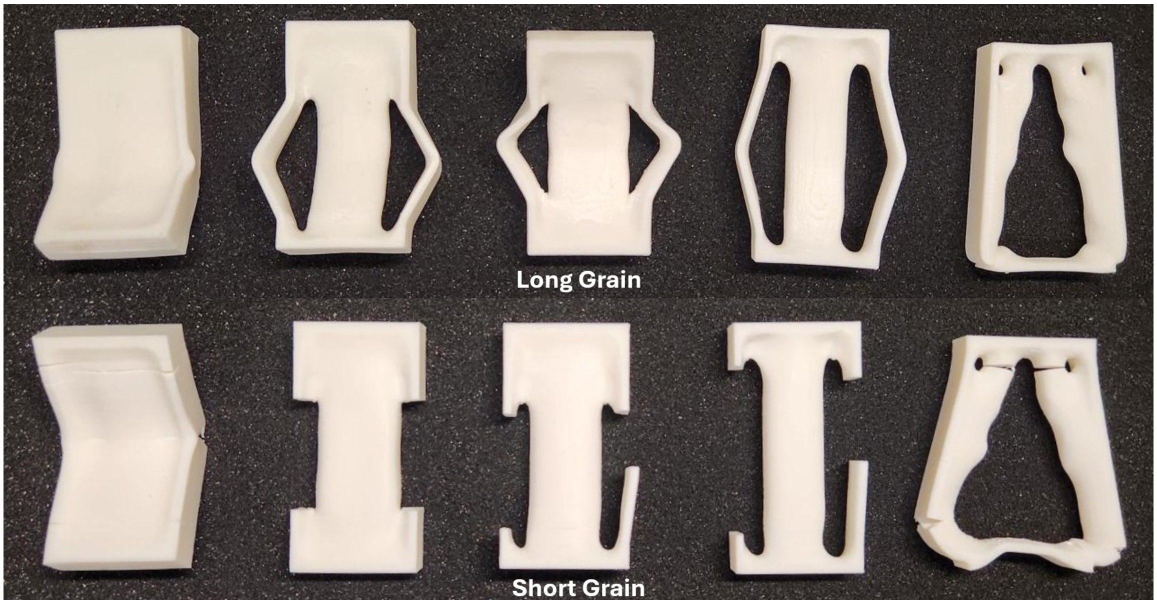

Figure 13: Failure mode for the 10 specimens (lined up from left to right in increasing mass reduction order from 30% to 70% reduction).

4 Discussion

4.1 Force-Displacement Behavior

Fig. 10 shows that the FEA model and the physical data agree well until the specimens begin to yield. This is expected since the FEA model assumes linear elastic behavior and does not capture geometric or material nonlinearities. This simplification was chosen due to the increased complexity of implementing nonlinear analysis for STL-based meshes.

Notably, the FEA model is able to reasonably predict failure for the 30% mass reduction case, as observed in both long- and short-grain specimens. However, for higher mass reductions, the discrepancy increases as failure becomes dominated by instability and material anisotropy rather than purely elastic behavior.

As mass reduction increases, the structural response becomes increasingly governed by slender load paths, which are more susceptible to local buckling. This explains why the deviation between FEA and experimental results grows with increasing mass reduction. The simplified linear model does not capture these instability-driven failure modes, leading to overprediction of strength.

Additionally, Fig. 11 shows that the maximum load decreases approximately linearly for long-grain specimens, whereas short-grain specimens exhibit a more rapid, nonlinear degradation in strength. This highlights the strong influence of manufacturing-induced anisotropy in FDM components.

4.2 FEA Stress Analysis

One advantage of the FEA model over physical testing is the ability to evaluate stress distributions, which are difficult to obtain experimentally for complex geometries. Since the FEA model shows good agreement with experimental results up to the design load of 3500 N, it is reasonable to use it for stress analysis within this range.

Fig. 12 shows the average Von Mises stress for each specimen, along with safety factors relative to a yield strength of 35 MPa. Based on a target safety factor of 2.0, the results suggest that approximately 50–60% mass reduction is acceptable for this loading condition.

However, it is important to note that the use of average stress may mask localized stress concentrations, particularly near thin members and junctions created by topology optimization. These localized regions are likely the initiation points for failure, meaning the true safety margin may be lower than indicated by the average stress values.

Furthermore, the assumption of isotropic material behavior in the FEA model neglects the directional dependence introduced by FDM printing. This is especially significant for short-grain specimens, where interlayer bonding strength governs failure. As a result, safety factors derived from FEA should be interpreted as optimistic estimates for printed components.

4.3 Failure Mode

As shown in Fig. 13, all specimens exhibit buckling as the primary failure mode due to their slender geometry. In addition, short-grain specimens experience significant cracking caused by interlayer delamination. This failure mechanism is characteristic of FDM-printed materials, where bonding between layers is weaker than within layers.

The topology-optimized designs introduce thin structural members that are highly sensitive to both buckling and material defects. In short-grain orientations, these members fail prematurely due to insufficient interlayer strength, leading to a more brittle and sudden failure compared to long-grain specimens.

4.4 Limitations and Sources of Error

Several limitations in this study contribute to discrepancies between the FEA predictions and experimental results:

- **Linear FEA assumptions:** The simulations assume linear elastic behavior and small deflections. As a result, geometric nonlinearities such as large deflections and buckling are not captured, which become significant at higher loads and for higher mass reductions.
- **Material modeling:** The PLA material is modeled as isotropic in the FEA, whereas FDM-printed components exhibit strong anisotropy depending on layer orientation. This leads to overprediction of strength, particularly for short-grain specimens.
- **Mesh representation:** The use of STL-based tetrahedral meshes introduces geometric discretization errors and may not accurately capture smooth stress gradients, especially in thin members and highly optimized regions.
- **Manufacturing variability:** Variations in print quality, layer adhesion, and minor defects can significantly influence failure behavior, particularly in slender features produced by topology optimization.

These limitations indicate that while the FEA model is effective for capturing general trends and relative performance, it should be used with caution when predicting absolute failure loads for topology-optimized, additively manufactured structures.

4.5 Design Implications

The results demonstrate that topology optimization can significantly reduce mass while maintaining acceptable structural performance up to a threshold of approximately 50–60% mass reduction for the loading conditions considered in this study. Beyond this point, structural performance degrades rapidly due to increased susceptibility to buckling and the presence of slender load paths.

Additionally, the results highlight the importance of incorporating manufacturing considerations into the design process. While topology optimization produces geometries that are optimal under idealized, isotropic assumptions, these designs may perform poorly when fabricated using anisotropic processes such as fused deposition modeling (FDM). In particular, layer orientation plays a critical role in determining failure behavior.

Future work should focus on integrating manufacturing constraints and material anisotropy directly into the topology optimization process. Doing so would enable the generation of designs that are not only optimal in simulation but also robust and reliable when physically realized.